

Forecasting Hays County Housing Market Speed

A One-Month-Ahead Decision Support Model

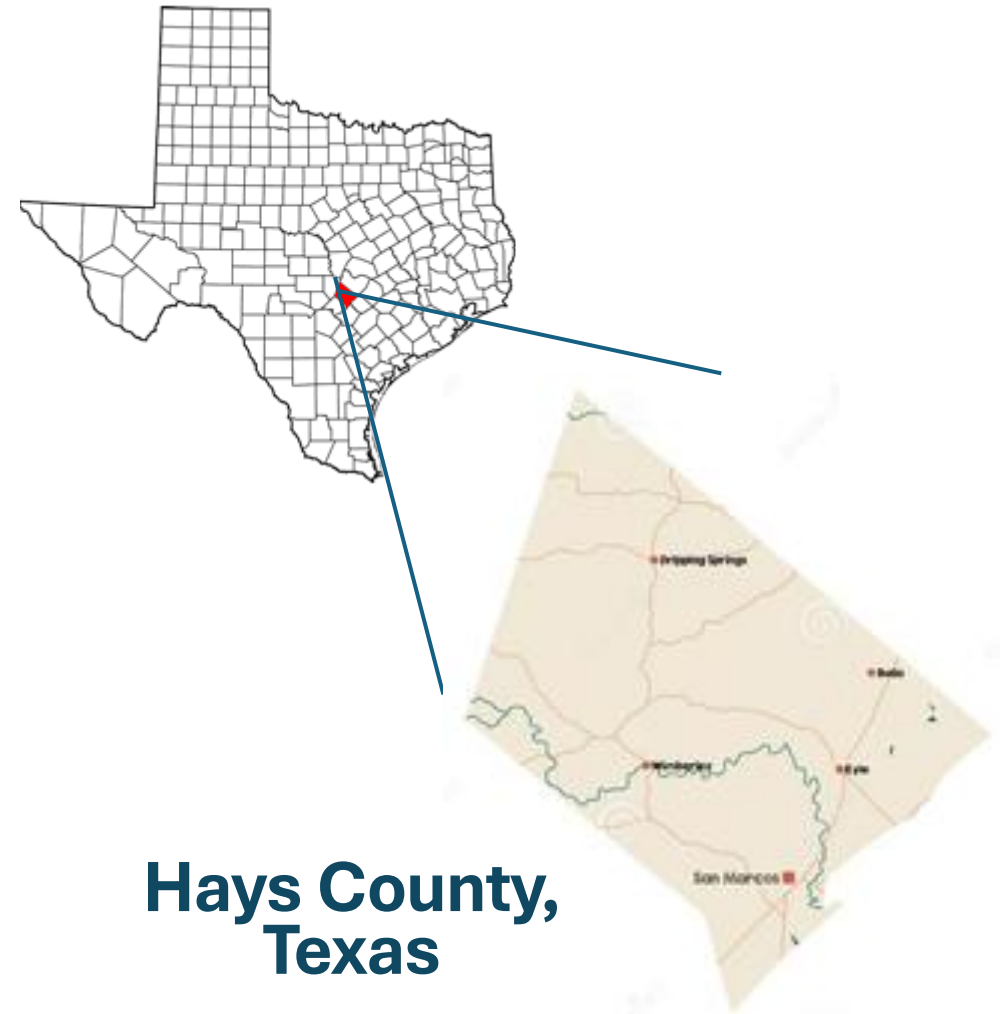
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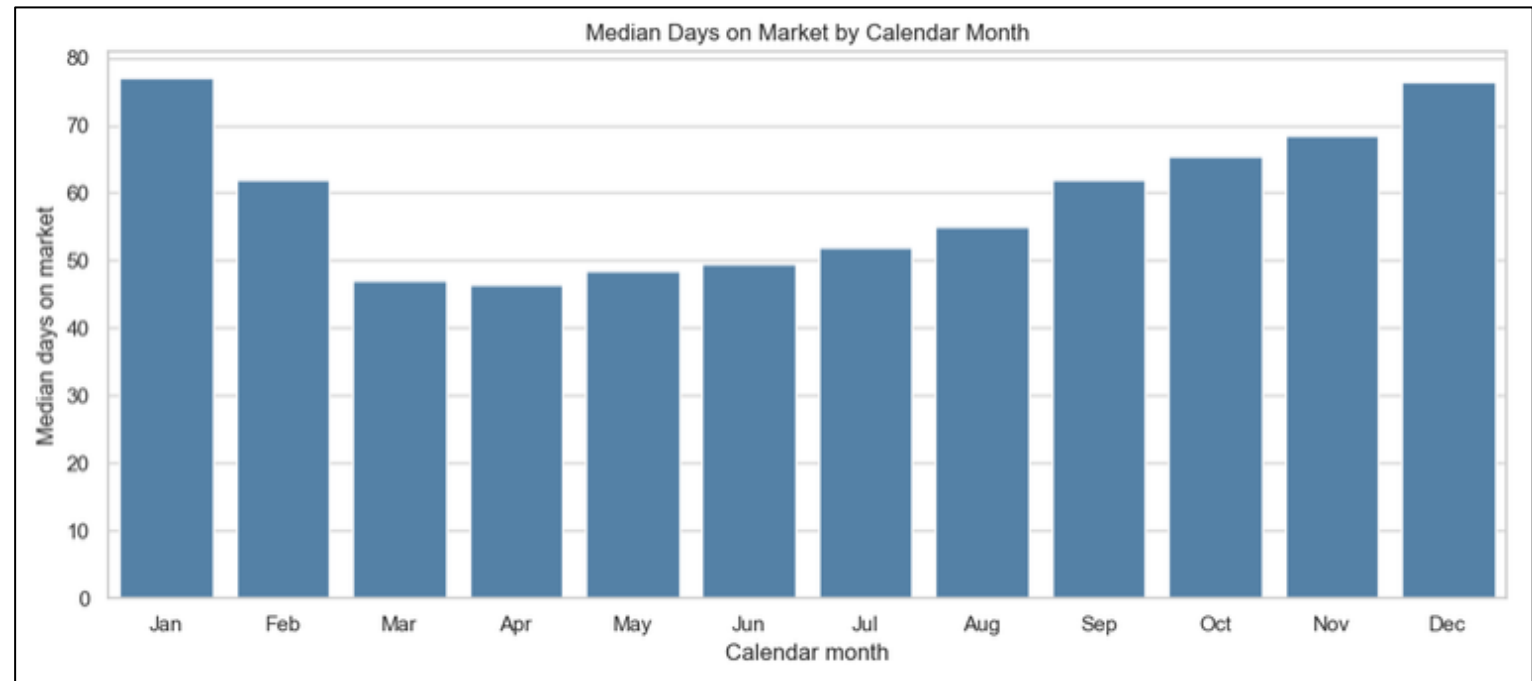
The Business Question

- Can we anticipate next month's market pace?
 - Outcome: countywide median days on market (MDOM)
 - Audience: real estate professionals advising buyers and sellers



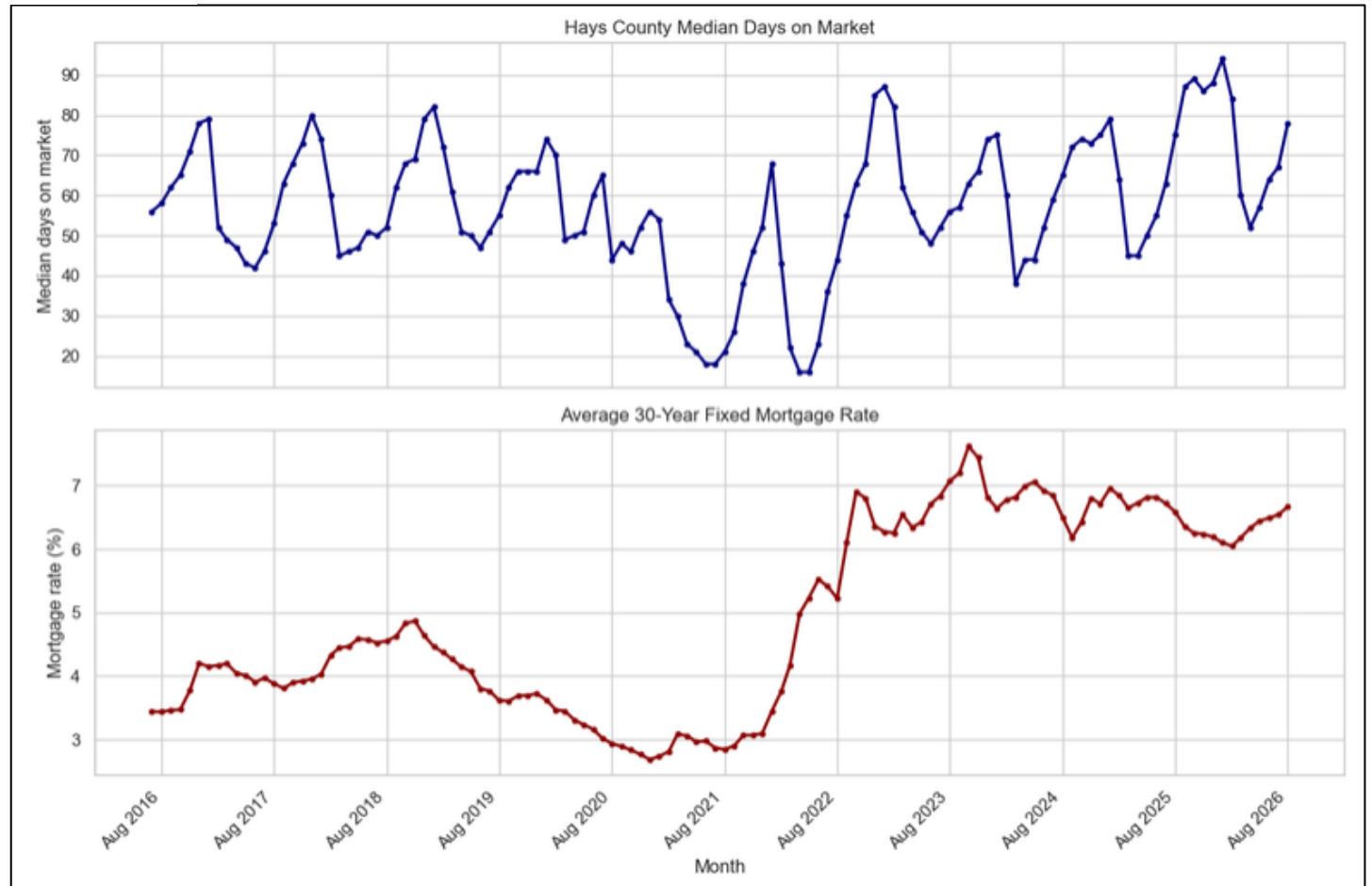
Why Market Speed Matters

- Market speed changes over time
 - January median: 77 Days
 - April median: 46.5 days



Data and Preparation

- Two sources, one monthly dataset
 - Realtor.com:
 - Hays County listings
 - Freddie Mac:
 - 30-year mortgage rates
 - July 2016 – August 2026, 122 months



A Fair Test of the Forecast

- Earlier months trained the models; later months tested them
 - Training: 97 Observations
 - Test: 24 Observations
 - Compared with previous month and prior-year forecasts

Training – 97 Observations
Aug 2016 – Aug 2024

Testing – 24
Observations
*Sep 2024 – Aug
2026*

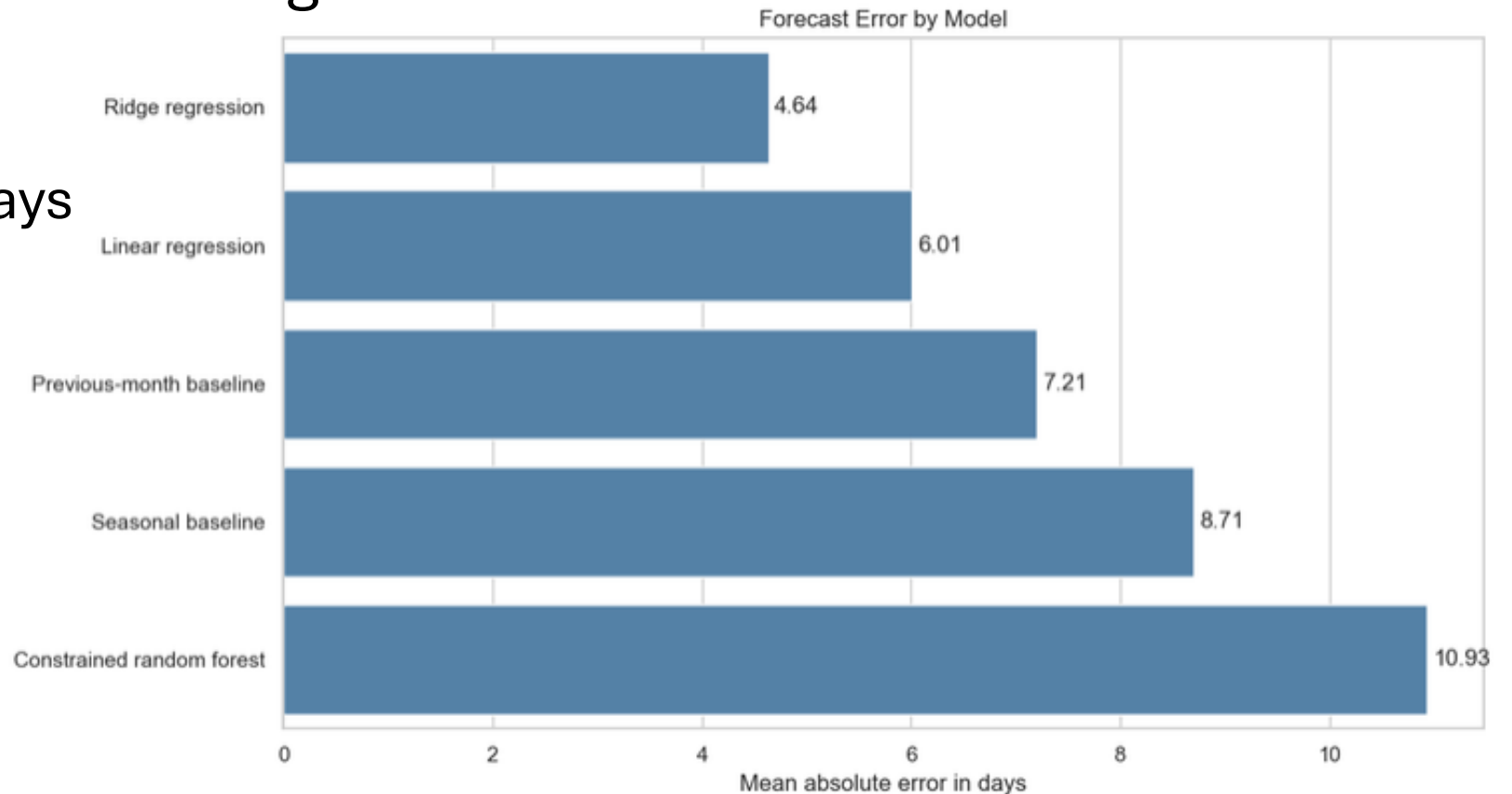
Forecast
Sep 2026

Model tuning used time-series cross-validation within the training period

The final 24 months were held back to test predictions on later data

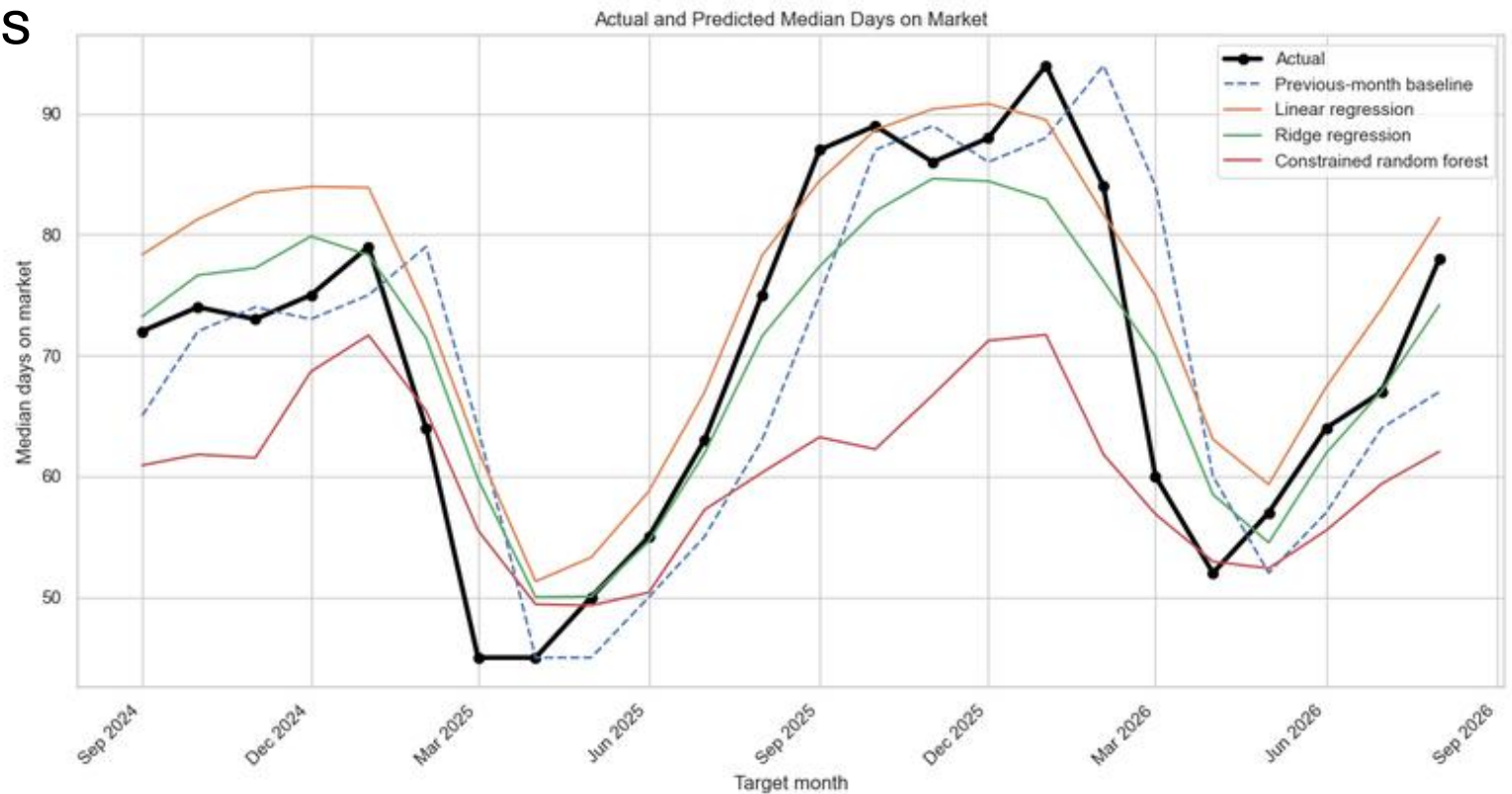
Which Approach Worked Best

- Ridge had the lowest average error
 - Ridge: 4.64 days
 - Previous-month baseline: 7.21 days
 - 35.7 % lower average error



What did the Predictions Look Like

- Ridge generally followed the observed changes
 - Test period: Sep 2024 – Aug 2026
 - Largest miss: about 14.7 days (Mar 2025)



September 2026 Estimate

- September Forecast: 82.5 days

78 Days

August Actual Days on Market



82.5 Days

September Estimated Days on Market

About 4.5 Days Longer

Countywide estimate; test-period average error: 4.64 days

What Informed the Forecast

- Current market speed and active listings mattered most
 - Other inputs:
 - New listings
 - Seasonality
 - Pending ratio
 - Prices
 - Mortgage rates

	Predictor	Coefficient	Absolute coefficient
0	Current median days on market	5.264	5.264
1	Active listings	4.524	4.524
2	New listings	-3.530	3.530
3	Seasonal sine term	-3.084	3.084
4	Seasonal cosine term	2.921	2.921
5	Pending listing ratio	-2.321	2.321
6	Median listing price	-1.359	1.359
7	Listings with price reductions	1.090	1.090
8	Pending listings	0.553	0.553
9	Average 30 year mortgage rate	0.158	0.158

Limits and Responsible Use

- Use the forecast with context
 - One county and one 24-month test period
 - Individual homes can differ
 - Show the historical error with every estimate
- An estimate is not a guarantee; do not use it to target or exclude protected groups

Recommendations

- Add a monthly forecast to a market dashboard showing:
 - Forecast
 - Latest actual
 - Expected change
 - Historical error
- Review unusual changes before sharing

***Ridge provided a useful countywide early indicator,
with an average test error of 4.64 days***

References

- Freddie Mac. (n.d.). Primary Mortgage Market Survey [Data set]. Retrieved September 5, 2026, from <https://www.freddiemac.com/pmms>
- Realtor.com Economic Research. (n.d.). Residential real estate data library [Data set]. Retrieved September 5, 2026, from <https://www.realtor.com/research/data/>
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